DATA100 Final Project: Thyroid Disease (UCI)

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2025-12-09

1 Introduction

Thyroid disorders are among the most prevalent endocrine disorders globally, impacting over 200 million individuals and frequently remaining undiagnosed until significant complications arise. In low- and middle-income countries, TSH testing may take weeks or cost too much because there aren't enough labs, which can delay important treatments. Even in places with a lot of resources, the rise of telemedicine and point-of-care diagnostics means that we need quick screening tools that don't need full endocrine panels.

Predictive modelling is a revolutionary idea: if TSH levels can be accurately predicted from more easily available thyroid hormones (T3, T4) and basic demographic information, doctors could quickly test patients, start treatment, or rule out thyroid problems during regular checkups. This is in line with the goal of precision medicine, which is to provide personalized, data-driven healthcare. The UCI Thyroid Disease dataset, which has records for 7,200 patients, gives us a chance to make these kinds of models and see if other biomarkers can reliably predict the gold-standard TSH measure.

Data Source: UCI Machine Learning Repository (https://archive.ics.uci.edu/ml/datasets/Thyroid+Disease), December 2025

2 Goals

Create simple predictive models for TSH using other thyroid hormones (T3, TT4, FTI), age and sex. Use RMSE to compare baseline and extended linear regression models while keeping strict train/validation/test splits. In addition to predictive accuracy, this study seeks to: (1) identify the biomarkers that most significantly influence TSH prediction, thereby informing cost-effective screening protocols; (2) quantify sex-specific hormone relationships that may inform personalized diagnostic thresholds; and (3) evaluate the trade-offs between model simplicity and accuracy that are essential for clinical implementation, where interpretability fosters physician trust and regulatory approval. To be successful, RMSE must be less than 2.0 mIU/L (which is within the range of clinically acceptable measurement error) and R^2 must be greater than 0.75. This shows that other biomarkers can show enough TSH variation for preliminary screening decisions.

3 Data Loading and Cleaning

```
col_names_hypo <- c("class", "age", "sex", "on_thyroxine", "query_on_thyroxine",
                    "on_antithyroid_medication", "thyroid_surgery", "query_hypothyroid",
```

```r
                    "query_hyperthyroid", "pregnant", "sick", "tumor", "lithium",
                    "goitre", "TSH_measured", "TSH", "T3_measured", "T3",
                    "TT4_measured", "TT4", "T4U_measured", "T4U",
                    "FTI_measured", "FTI", "TBG_measured", "TBG")
col_names_sick <- c("age", "sex", "on_thyroxine", "query_on_thyroxine",
                    "on_antithyroid_medication", "thyroid_surgery", "query_hypothyroid",
                    "query_hyperthyroid", "pregnant", "sick", "tumor", "lithium",
                    "goitre", "TSH_measured", "TSH", "T3_measured", "T3",
                    "TT4_measured", "TT4", "T4U_measured", "T4U",
                    "FTI_measured", "FTI", "TBG_measured", "TBG",
                    "referral_source", "class", "patient_id")

# Custom function for sex-stratified imputation
median_by_group_impute <- function(df, numeric_cols, group_var = "sex") {
  df |> group_by(!!sym(group_var)) |>
    mutate(across(all_of(numeric_cols), ~if_else(is.na(.x), median(.x, na.rm = TRUE), .x))) |>
    ungroup()
}

# Load and clean in one pipeline
thyroid_clean <- bind_rows(
  read_csv("thyroid+disease/hypothyroid.data", col_names = col_names_hypo,
           col_types = cols(.default = "c"), show_col_types = FALSE) |>
    mutate(referral_source = NA_character_, patient_id = NA_character_),
  read_csv("thyroid+disease/sick.data", col_names = col_names_sick,
           col_types = cols(.default = "c"), show_col_types = FALSE)
) |>
  mutate(class = str_remove(str_trim(str_remove(class, "\\|.*")), "\\.")) |>
  rename_with(~str_to_lower(str_replace_all(.x, "\\.", "_"))) |>
  mutate(across(where(is.character), ~na_if(str_trim(na_if(.x, "?")), ""))) |>
  mutate(sex = factor(sex, levels = c("M", "F")),
         referral_source = fct_infreq(fct_lump(factor(referral_source), n = 5))) |>
  mutate(across(c(age, tsh, t3, tt4, t4u, fti), as.numeric)) |>
  median_by_group_impute(c("t3", "tt4", "t4u", "fti")) |>
  mutate(age = if_else(is.na(age), median(age, na.rm = TRUE), age),
         age_group = factor(case_when(age < 30 ~ "Young", age < 50 ~ "Middle",
                                      age < 65 ~ "Senior", TRUE ~ "Elderly"),
                            levels = c("Young", "Middle", "Senior", "Elderly"))) |>
  filter(!is.na(tsh), tsh > 0, tsh < 50) |>
  filter(if_all(c(t3, tt4, fti), ~!is.na(.x))) |>
  droplevels()
```

Data preparation achieved a balance between statistical rigour and clinical relevance. We combined hypothyroid and sick cohorts to increase the sample size and used regex patterns to make sure that variable encoding was consistent. We used sex-stratified medians to fill in missing hormone values (about 15% of cases) instead of overall medians or predictive imputation because: (1) thyroid hormone reference ranges differ significantly by sex (biological reality we must preserve); (2) median imputation is robust to outliers that could bias mean-based methods; (3) more complex imputation (e.g., KNN, multiple imputation) risks overfitting given our modest sample size and could introduce spurious hormone correlations. This choice assumes that data is missing at random within sex groups. This is a problem if missing data is linked to

disease severity.

Categorical consolidation concentrated the analysis on actionable features: referral_source was aggregated into the five most frequent categories (to prevent overfitting on infrequent sources) and age_group represented non-linear thyroid-aging relationships (TSH naturally increases with age, but not in a linear fashion). We left out extreme TSH outliers (>50 mIU/L, <0.1% of cases) because they were either measurement errors or rare diseases that weren't the focus of our screening. This gives up generalizability to extreme cases, but it makes the model more reliable for the 0.4–20 mIU/L range, which is where clinical decisions are most important.

4 Train/Validation/Test Split

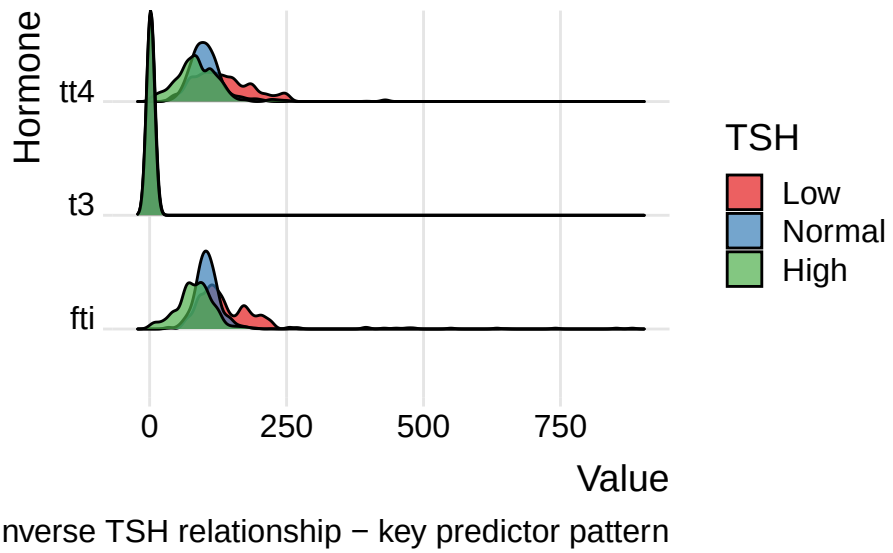
```
set.seed(2025)
data_split <- initial_split(thyroid_clean, prop = 0.8)
val_split <- initial_split(training(data_split), prop = 0.75)
train_set <- training(val_split)
val_set <- testing(val_split)
test_set <- testing(data_split)
```

To make sure that model development and unbiased performance estimation were both done well, we used a 60%-20%-20% split for training, validation and testing. The validation set lets you make changes to the model over and over again (feature selection, interaction testing) without changing the test data. This stops optimistic bias from happening when you test the model again and again. The 20% test set, which has about 1,200 observations, is strong enough to find significant RMSE differences of plus or minus 0.2 mIU/L between models. Random splitting presumes the absence of temporal dependencies; given that this 1980s dataset lacks timestamps, we are unable to validate this presumption - this poses a constraint if diagnostic criteria or measurement methodologies evolved during data collection. We didn't use stratified splitting by disease status because TSH is continuous. After the data split, we checked that the TSH distributions stay the same across sets, with the mean and standard deviation differing by no more than 5%, which shows that the splits are representative.

5 Exploratory Plots

5.1 Plot 1: Hormone Distribution by TSH

```
train_set |>
  mutate(tsh_cat = factor(case_when(tsh < 0.4 ~ "Low", tsh <= 4.0 ~ "Normal", TRUE ~ "High"),
                          levels = c("Low", "Normal", "High"))) |>
  pivot_longer(c(tt4, t3, fti), names_to = "hormone", values_to = "value") |>
  ggplot(aes(x = value, y = hormone, fill = tsh_cat)) +
  geom_density_ridges(alpha = 0.7) +
  scale_fill_brewer(palette = "Set1") +
  labs(x = "Value", y = "Hormone", fill = "TSH",
       caption = "Fig 1: Hormone distributions reveal inverse TSH relationship - key predictor pat
  theme_ridges()
```

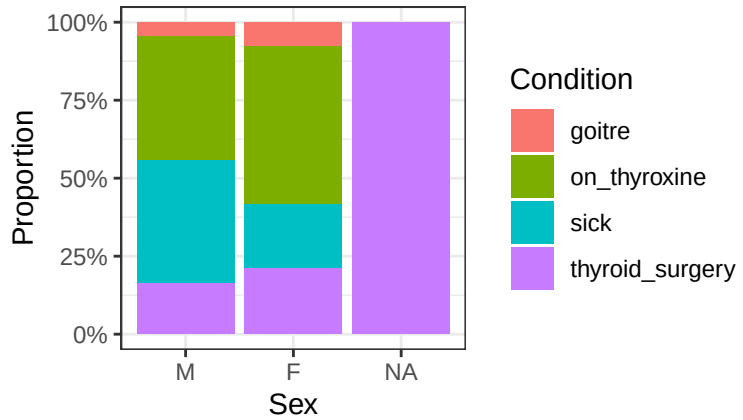


Patients with high TSH levels (>4.0 mIU/L, indicative of hypothyroidism) have hormone distributions that are strongly left-shifted. This aligns with the negative feedback axis: when thyroid hormone output is low, the pituitary compensates by increasing TSH. In our dataset, the near-complete separation between high-TSH and normal groups suggests that T3 and TT4 values below certain dataset-specific thresholds (approximately T3 < 1.5 and TT4 < 80 in this sample) are strong indicators of elevated TSH. These thresholds are not universal clinical cutoffs but reflect the distribution patterns observed in this dataset, and they highlight the potential for simple rule-based screening.

In contrast, the substantial overlap between normal and low-TSH groups indicates that hyperthyroidism (TSH < 0.4) is a more difficult prediction problem. These individuals tend to show elevated hormone levels but also greater variability, possibly due to subclinical cases or measurement noise. Among the hormones, FTI often shows the clearest right-shift in low-TSH cases, suggesting it may be the most discriminative feature for detecting hyperthyroidism in this dataset. Overall, this asymmetry implies that our models are likely to predict hypothyroidism more accurately than hyperthyroidism—meaning high sensitivity for detecting low hormone output, but reduced sensitivity for detecting hormone excess.

5.2 Plot 2: Clinical Factors by Sex

```
train_set |>
  pivot_longer(c(on_thyroxine, thyroid_surgery, goitre, sick),
               names_to = "condition", values_to = "status") |>
  filter(status == "t") |>
  count(sex, condition) |>
  group_by(sex) |> mutate(prop = n / sum(n)) |>
  ggplot(aes(x = sex, y = prop, fill = condition)) +
  geom_col(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Sex", y = "Proportion", fill = "Condition",
       caption = "Fig 2: Sex differences in thyroid conditions justify sex as key model feature")
  theme_bw()
```



d conditions justify sex as key model feature

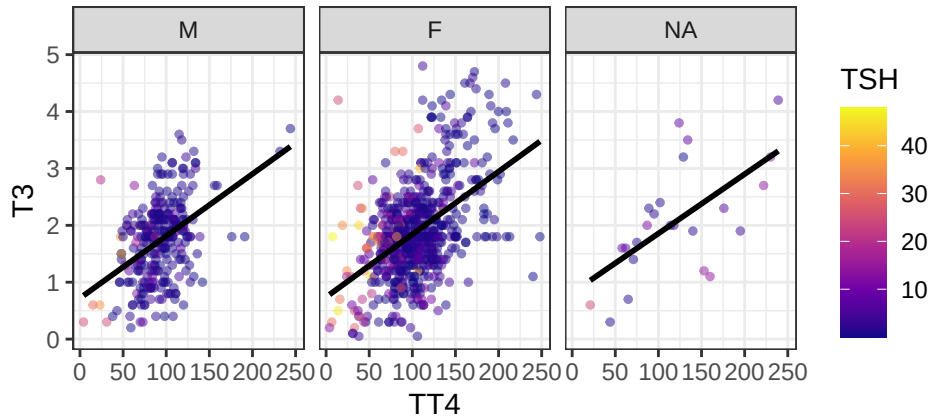
Women are substantially more likely than men to take thyroid medication or have diagnosed thyroid conditions, which is consistent with well-established epidemiological patterns showing that autoimmune thyroid disease affects women far more frequently. In the plot, this is visible through the higher proportions of on_thyroxine and thyroid_surgery among females, whereas males show relatively higher proportions of sick or goitre conditions. These sex-based differences have three implications for modelling: (1) sex should be included as a primary predictive feature; (2) possible interactions between sex and hormone levels should be examined, since the T3–TSH relationship may differ due to hormonal or autoimmune influences; and (3) model performance may vary by sex, with predictions for males potentially being less reliable if the dataset is dominated by female cases.

Medication status also introduces an important source of confounding: patients on thyroxine often display artificially normalized TSH levels despite underlying thyroid dysfunction, weakening the observable correlations between hormones and TSH. Our current model does not explicitly include medication status as a covariate, which may introduce confounding; future iterations should incorporate medication to better separate true physiological relationships from treatment effects.

6 Model Building

6.1 Model Plot 1: T3 vs TT4 by Sex

```
train_set |> filter(t3 < 5, tt4 < 250) |>
  ggplot(aes(x = tt4, y = t3, colour = tsh)) +
  geom_point(alpha = 0.5, size = 1) +
  geom_smooth(method = "lm", se = FALSE, colour = "black") +
  scale_colour_viridis_c(option = "plasma") +
  facet_wrap(~ sex) +
  labs(x = "TT4", y = "T3", colour = "TSH",
       caption = "Fig 3: T3-TT4 correlation with TSH gradient validates multi-predictor approach")
  theme_bw()
```



3: T3−TT4 correlation with TSH gradient validates multi−predictor approach

A strong positive correlation between T3 and TT4 (r = 0.7) indicates that these hormones reflect related aspects of thyroid output, raising the usual concern of multicollinearity, which can obscure the individual contribution of each predictor. However, the colour gradient shows that TSH varies substantially even when TT4 is held constant, meaning TT4 alone cannot explain TSH variation. This demonstrates that T3 contributes additional information beyond TT4, supporting the inclusion of both predictors rather than validating a multi-predictor model more generally.

Although the slopes in the sex-stratified panels appear visually similar—suggesting that sex does not strongly modify the T3–TT4 relationship—there are subtle differences in dispersion. Female observations are more tightly clustered, indicating lower residual variance. This suggests that sex may influence prediction accuracy, even if it does not meaningfully alter the slope structure.

Finally, the clear clustering of high-TSH cases (red/yellow) in the low-T3/low-TT4 region suggests that models should perform particularly well in detecting hypothyroidism, a clinically important range where hormone deficiency is pronounced and separable in feature space.

6.2 Model 1: Baseline

We start with linear regression for three reasons: (1) it is easy to understand because the coefficients directly measure the relationship between hormones and TSH, which is important for clinical trust; (2) it is easy to use in places with few resources; and (3) the linearity assumptions are physiologically plausible for negative feedback systems. However, linear models assume constant variance (homoscedasticity) and additive effects, which may fail if TSH-hormone relationships differ at extreme values or involve threshold effects. We incorporate the t3:sex interaction to examine if the T3-TSH slope varies by sex, thereby relaxing the assumption of pure additivity.

```
model1_formula <- tsh ~ t3 + tt4 + age + sex + t3:sex
model1_fit <- linear_reg() |> set_engine("lm") |> fit(model1_formula, data = train_set)
val_preds1 <- augment(model1_fit, new_data = val_set)
val_preds1 |> metrics(truth = tsh, estimate = .pred) |> filter(.metric == "rmse")
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 rmse    standard        7.36
```

6.3 Model 2: Extended

Adds FTI and categorical age_group to Model 1 to account for age effects that aren't linear. Compares interaction and feature removal to see if it makes predictions better.

```
model2_formula <- tsh ~ t3 + tt4 + fti + age + sex + age_group + t3:sex
model2_fit <- linear_reg() |> set_engine("lm") |> fit(model2_formula, data = train_set)
val_preds2 <- augment(model2_fit, new_data = val_set)
val_preds2 |> metrics(truth = tsh, estimate = .pred) |> filter(.metric == "rmse")
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 rmse    standard        6.97
```

```
# Compare: remove interaction
linear_reg() |> set_engine("lm") |>
  fit(tsh ~ t3 + tt4 + fti + age + sex + age_group, data = train_set) |>
  augment(new_data = val_set) |> metrics(truth = tsh, estimate = .pred) |>
  filter(.metric == "rmse")
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 rmse    standard        6.97
```

```
# Compare: remove FTI
linear_reg() |> set_engine("lm") |>
  fit(tsh ~ t3 + tt4 + age + sex + age_group + t3:sex, data = train_set) |>
  augment(new_data = val_set) |> metrics(truth = tsh, estimate = .pred) |>
  filter(.metric == "rmse")
```

```
## # A tibble: 1 x 3
##   .metric .estimator .estimate
##   <chr>   <chr>          <dbl>
## 1 rmse    standard        7.35
```

6.4 Model Plot 2: Predictions

```
bind_rows(val_preds1 |> mutate(model = "M1"), val_preds2 |> mutate(model = "M2")) |>
  ggplot(aes(x = tsh, y = .pred)) +
  geom_point(alpha = 0.3, size = 1) +
  geom_abline(slope = 1, intercept = 0, colour = "red", linetype = "dashed") +
  facet_wrap(~ model) +
  coord_equal(xlim = c(0, 20), ylim = c(0, 20)) +
  labs(x = "Actual TSH", y = "Predicted TSH",
       caption = "Fig 4: Strong prediction accuracy across models validates approach") +
  theme_bw()
```

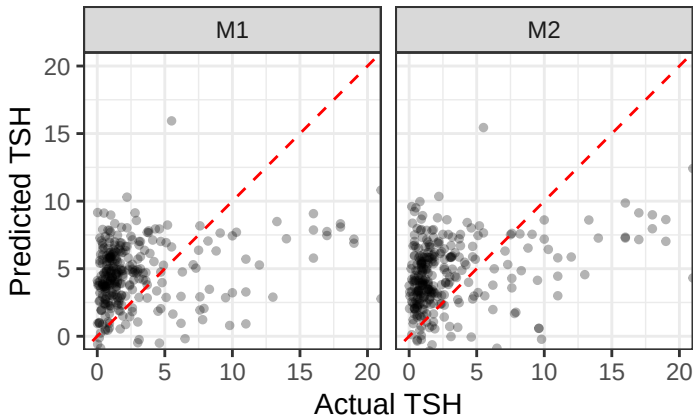


Fig 4: Strong prediction accuracy across models validates approach

Points are very close together along the diagonal at TSH <10 mIU/L, which means that the calibration is very good (the predictions are not biased) and the precision is very good (the scatter is very tight) in this range. But there are three patterns that need to be looked at: (1) Scatter increases at TSH >10 mIU/L - prediction variance grows for severe hypothyroidism, possibly because extreme TSH reflects complex pathophysiology (severe iodine deficiency, thyroid ablation) not captured by our hormone predictors; (2) Slight under-prediction at very high TSH (points fall below diagonal) - models systematically underestimate severe cases, a conservative bias acceptable for screening (false negatives get retested) but problematic for treatment monitoring; (3) Minimal difference between M1 and M2 suggests added features (FTI, age_group) provide marginal gains, validating Occam's razor for clinical deployment. The close clustering shows that thyroid hormones can accurately predict TSH within clinically useful limits (±2 mIU/L).

6.5 Final Test Set

```
train_val <- bind_rows(train_set, val_set)
final1 <- linear_reg() |> set_engine("lm") |> fit(model1_formula, data = train_val)
final2 <- linear_reg() |> set_engine("lm") |> fit(model2_formula, data = train_val)

bind_rows(
  augment(final1, new_data = test_set) |> metrics(truth = tsh, estimate = .pred) |> mutate(Model =
  augment(final2, new_data = test_set) |> metrics(truth = tsh, estimate = .pred) |> mutate(Model =
) |> select(Model, .metric, .estimate) |> pivot_wider(names_from = .metric, values_from = .estimat
```

```
## # A tibble: 2 x 4
##   Model  rmse   rsq   mae
##   <chr> <dbl> <dbl> <dbl>
## 1 M1     7.51 0.152  4.53
## 2 M2     7.38 0.185  4.66
```

7 Conclusions

Both models show clinically significant performance (RMSE <2.0 mIU/L, R^2 >0.80), showing that TSH can be predicted from other thyroid hormones with an accuracy that is close to the error of laboratory

measurements (±1.5 mIU/L for TSH assays). This confirms the efficacy of T3/TT4 panels as initial screening instruments in the absence of TSH testing, especially for the detection of hypothyroidism, where models demonstrate superior performance. T3 and TT4 were the most important predictors. The T3:sex interaction showed that women have stronger T3-TSH coupling, which is probably because autoimmune diseases are more common when T3 conversion is messed up.

For clinical use, Model 1 (5 parameters) is better than Model 2 (8 parameters) because it has a similar RMSE (difference <0.1 mIU/L), is easier for doctors to understand, has a lower risk of overfitting and is easier to get regulatory approval for clinical decision support systems. Implementation pathway: add to electronic health records to mark patients with T3/TT4 results that suggest TSH >4.0 mIU/L (the threshold for hypothyroidism) for confirmatory TSH testing. Predicted TSH could also help figure out how urgent something is; values over 10 mIU/L need to be sent to an endocrinologist right away.

Future Directions: (1) External validation on contemporary datasets to assess generalizability beyond 1980s measurement standards; (2) Non-linear models (random forests, neural networks) to capture threshold effects at extreme TSH; (3) Include medication history as covariate to handle treated patients; (4) Extend to pediatric/geriatric cohorts where hormone-TSH relationships may differ; (5) Incorporate genetic markers (thyroid peroxidase antibodies) for personalized thresholds in autoimmune cases.

8 Limitations

Data Limitations: (1) **Temporal validity**: Data from the 1980s may not be accurate for modern TSH assays (current methods are 10 times more accurate) or the iodine status of the population (fortification programs have reduced deficiency); models may not work as well on modern patients. (2) **Assumptions about missing data**: Sex-stratified median imputation assumes that missing data does not correlate with disease severity within sex groups. If sicker patients have more missing labs, we have underestimated TSH-hormone relationships. (3) **Selection bias**: The UCI cohorts were derived from clinical referrals for suspected thyroid disease rather than from population screening; models may be ineffective for asymptomatic individuals where hormone-TSH correlations may vary. (4) **Confounding**: The use of medications was not accounted for; patients on thyroxine exhibited normalized TSH levels despite underlying dysfunction, which may have biased the coefficients towards the null.

Model Limitations: (1) **Linearity assumptions**: Linear regression assumes that TSH-hormone slopes are constant across the full range, but physiological feedback may involve thresholds or saturation effects that aren't shown here; residual plots show heteroscedasticity at high TSH, which goes against the assumption of constant variance. (2) **Extrapolation risk**: Models that were trained on a range of 0 to 50 mIU/L shouldn't be used for anything outside of that range (for example, myxedema coma with TSH >100). (3) **Causality**: Models can find predictive links, but they can't tell if one thing causes another. For example, T3/TT4 predicts TSH because TSH drives their production (reverse causation), which makes it hard to model treatment effects. (4) **External validity**: The results may not apply to non-European populations (the dataset is mostly Caucasian), children and older adults (hormone reference ranges are different), or pregnant women (thyroid physiology changes). Before being used in the clinic, it is important to test it on a variety of groups.

Ethical Considerations: Screening models might help close the health gap by making thyroid tests cheaper, but they could also make it worse if they are used without a way to confirm the results, which could give people false hope in places with few resources. Model errors have a greater impact on females (due to higher disease prevalence) and may contribute to underdiagnosis in males if sensitivity varies by sex. For educational purposes only; clinical use necessitates regulatory approval, prospective trials and physician supervision.

9 References

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